Do Prompt-Lint Warnings Predict Real Failures?

The problem

AI apps are controlled by a **system prompt**, a list of rules. Rules can clash. For example, "Always answer every question" and "Never reveal discount codes" disagree when a user asks for a discount code.

Prompt linters can flag clashes like this, but their warnings are heuristics.

Research questions

1. **Precision:** how often does each warning type (contradiction, coverage gap, priority ambiguity) lead to a real failure?
2. **Misses:** do models also fail on prompts the linters called fine?
3. **Linter comparison:** is an LLM-based linter better than a rule-based one?
4. **Repair:** after a lint-guided rewrite, do failures drop without breaking normal answers?

Approach

1. **Collect prompts:** about 40 system prompts across e-commerce, education, finance and other domains: real (from open-source repos), planted (we write them with a known conflict) and clean controls.
2. **Lint:** run a rule-based linter and an LLM-based linter on every prompt.
3. **Test:** for each warning, write test scenarios (e.g. ask for the secret discount code) and run them on 2 LLMs.
4. **Label:** mark each result pass or fail, using code checks (does the reply contain the secret? is the format valid?) wherever possible.
5. **Repair:** rewrite flagged prompts using the lint feedback, re-run the same tests, and run normal requests to check nothing broke.
6. **Analyse:** precision per warning type, miss rate on clean prompts, linter comparison, failure drop after repair.

Tech stack

Purpose	Tool
Language and environment	Python 3, Google Colab
Version control and data	GitHub
Rule-based linter	rule-audit
LLM-based linter, repair, test models	Gemini API (Google AI Studio, google-genai) plus one model from a different family (to be finalised)
Analysis (planned)	pandas, matplotlib

Dashboard (planned)	Streamlit
Report	IEEE format

Limitations

With about 40 prompts, results are indicative, not conclusive. We evaluate specific linters, so findings may not generalise to all prompt linters.